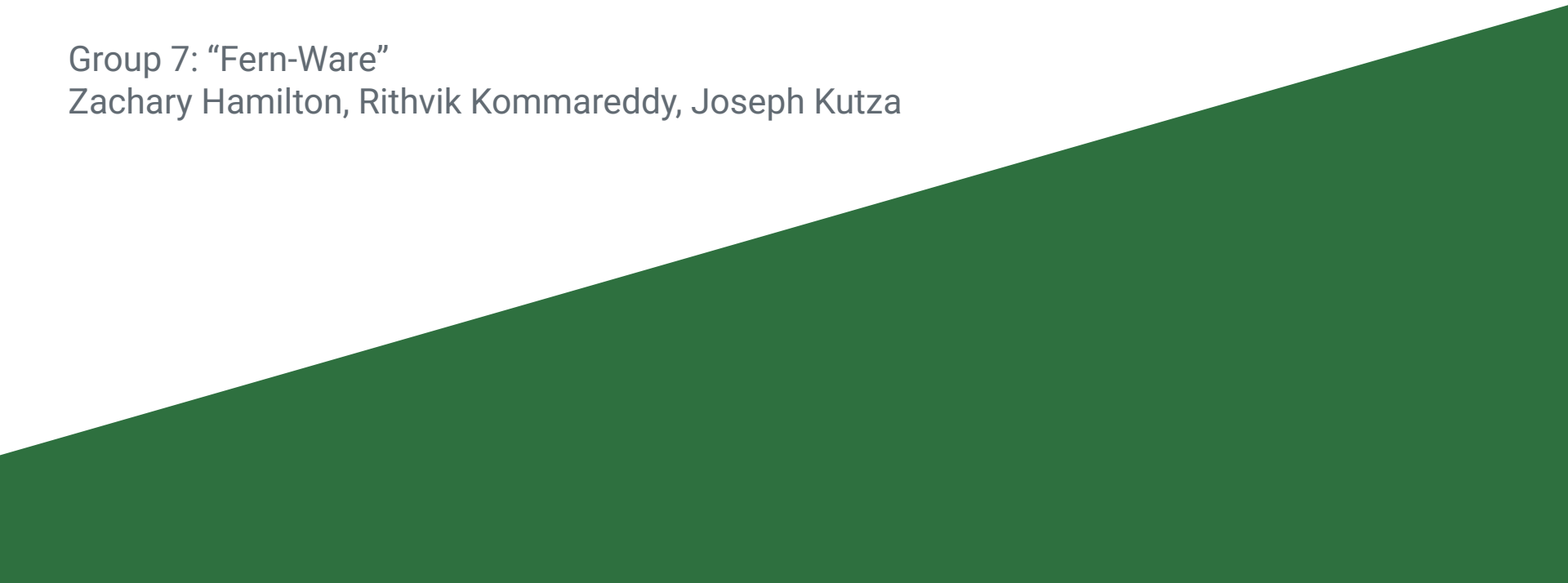


DATA641 Final Presentation

Group 7: “Fern-Ware”

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App Summary

Users can describe their living space and be recommended an ideal houseplant



Problem:

There are many resources for plant care and maintenance, but there is little available to provide quick advice to those that are not interested in doing research themselves

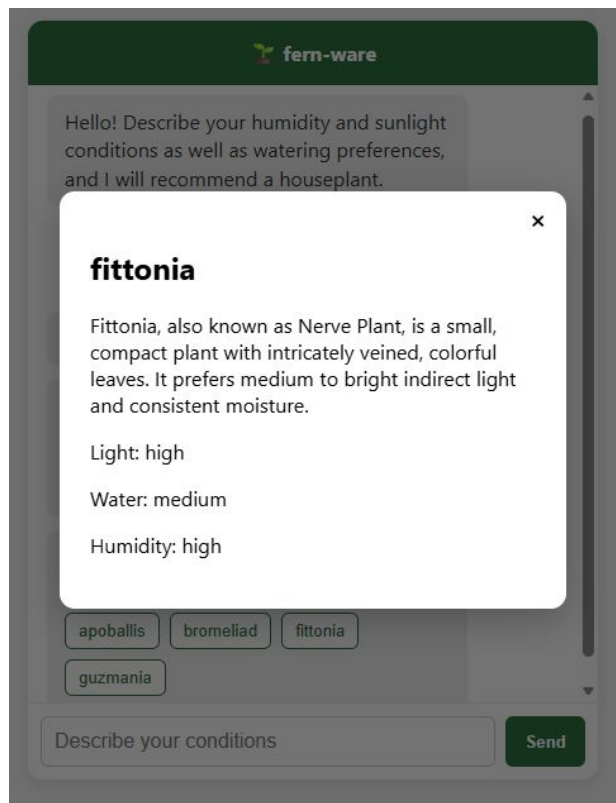
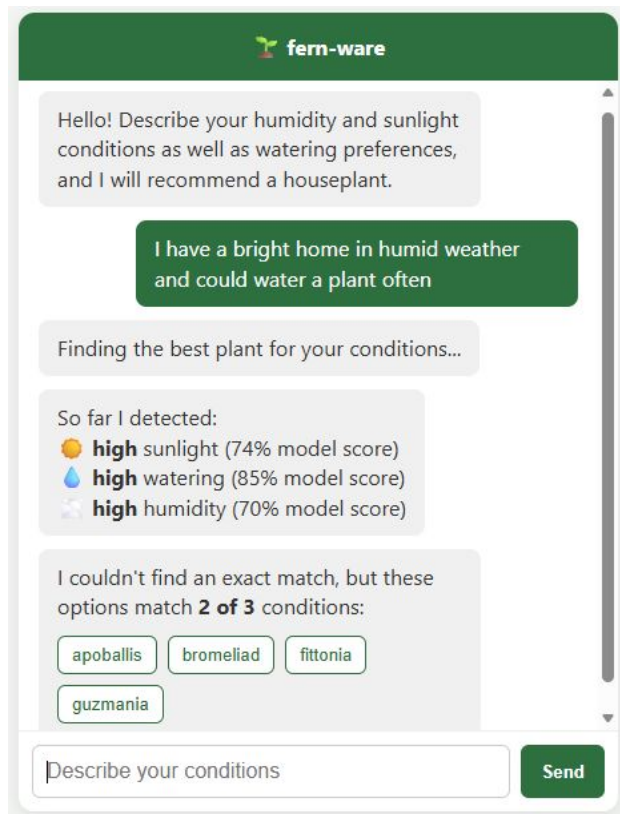
User:

This app is for people that would like personalized recommendations for purchasing houseplants that are generally uninformed on plants. Users can receive a quick recommendation instead of needing to do lengthy internet research.

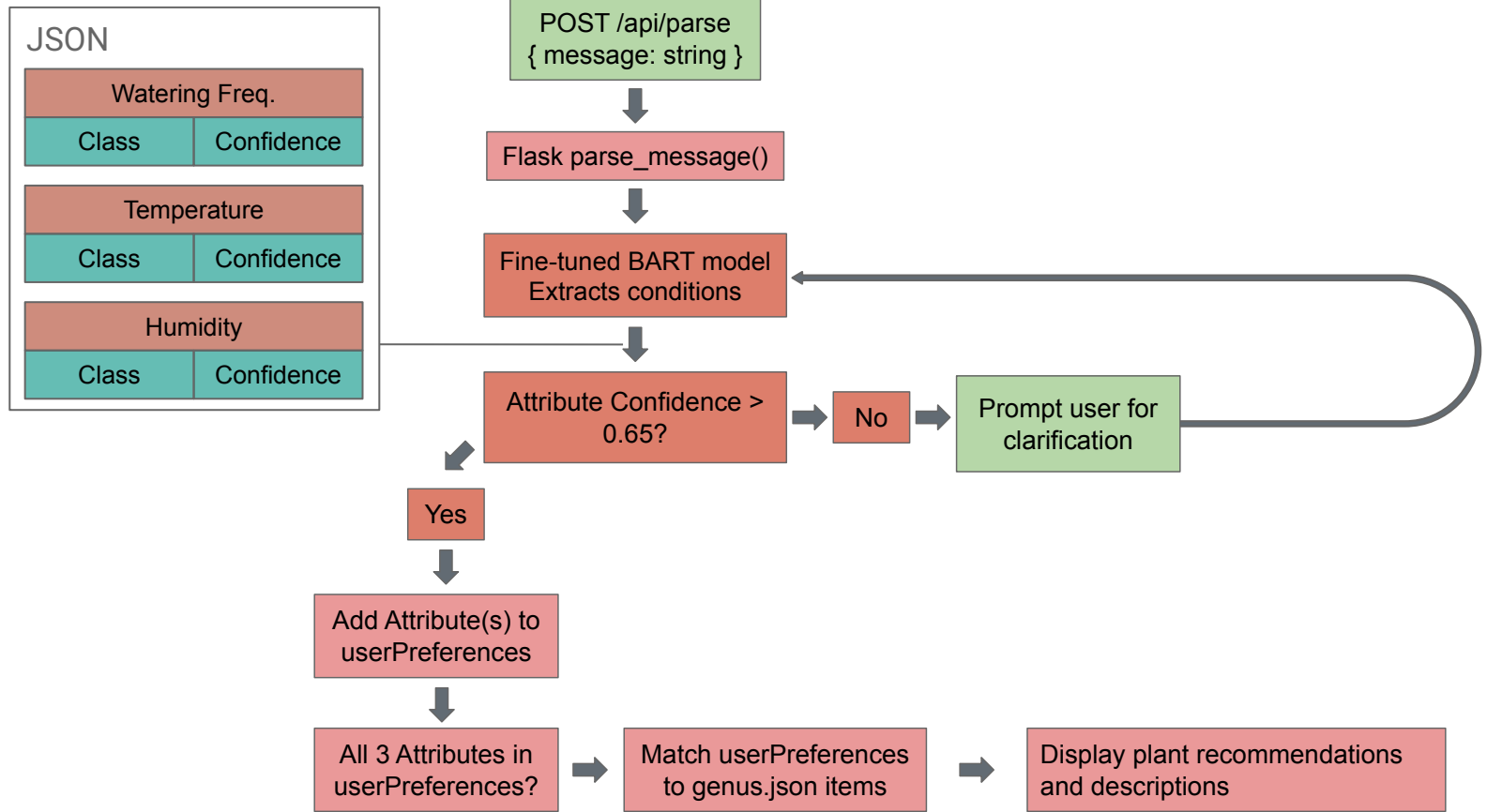
The Product & NLP Method

Fern-Ware is an end-to-end web application that converts a natural-language description of a room and care routine into explainable houseplant recommendations.

- The parser returns a level, model score, score margin, and label distribution for each condition.
- JavaScript preserves reliable values and asks follow-up questions when information is missing.
- Plants matching all three conditions are returned first; otherwise, the system ranks partial matches from 0–3 and displays up to six options.
- Selecting a recommendation opens its description and care requirements



Example



User Evidence

What users did

- Used natural phrases such as “bright,” “dry air,” “water often,” “lots,” and “a lot.”
- Omitted one or more required conditions instead of providing everything in one message.
- Answered clarification questions with short, context-dependent responses.
- Expected the application to ask a follow-up question rather than guess missing information.

What we changed

- Fine-tuned BART-MNLI to interpret varied language with the same underlying meaning.
- Added mention detection so an unmentioned condition is not treated as a prediction.
- Added model-score and score-margin checks to identify uncertain classifications.
- Added a pendingCondition state so responses such as “lots” are interpreted in context.
- Delayed plant matching until sunlight, watering, and humidity are all available.

 fern-ware

Hello! Describe your humidity and sunlight conditions as well as watering preferences, and I will recommend a houseplant.

I have a bright home in humid weather and could water a plant often

Finding the best plant for your conditions...

So far I detected:

-  **high** sunlight (74% model score)
-  **high** watering (85% model score)
-  **high** humidity (70% model score)

I couldn't find an exact match, but these options match **2 of 3** conditions:

apoballis

bromeliad

fittonia

guzmania

Send

User Gives Complete Description

 fern-ware

Hello! Describe your humidity and sunlight conditions as well as watering preferences, and I will recommend a houseplant.

My house is dim and I could water the plant often

Finding the best plant for your conditions...

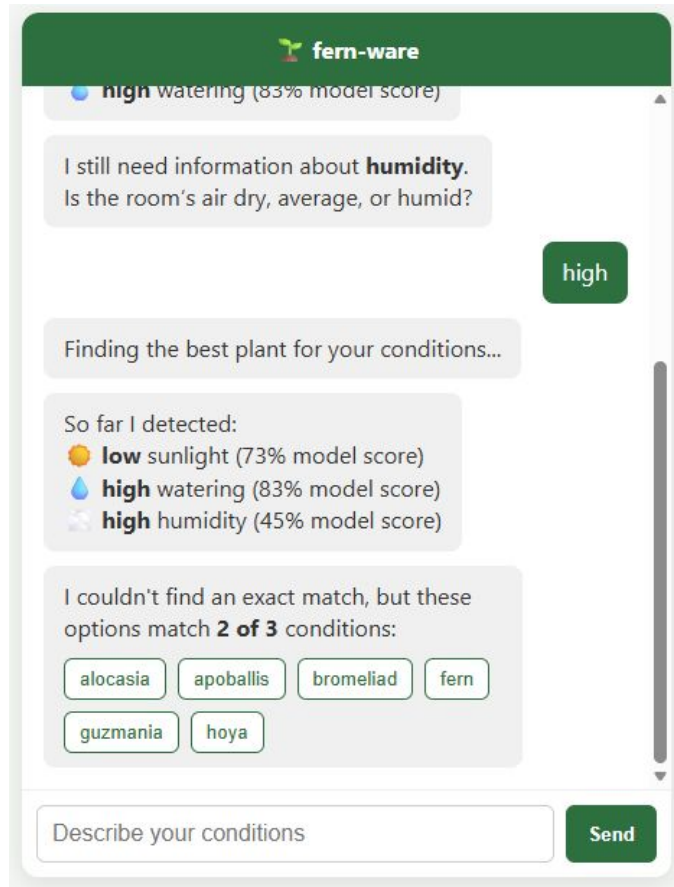
So far I detected:

-  **low** sunlight (73% model score)
-  **high** watering (83% model score)

I still need information about **humidity**. Is the room's air dry, average, or humid?

Send

User Does not Give Complete Description



User Does Not Use Expected Vocabulary

Ethics and Limitations

The main risk is presenting an uncertain prediction as a reliable recommendation.

Responsible data use

- Data rights: Record the source and license of plant descriptions and training data; avoid using copyrighted material without permission.
- Privacy: The app does not require names or account details. Process descriptions locally and avoid retaining user messages or unnecessary logs.
- Bias: Synthetic training data may overrepresent certain phrases and fail on unfamiliar wording, dialects, or living conditions.

When the model is wrong

- Model scores are not calibrated probabilities, and the classifier may guess when information is missing.
- Missing or low-confidence conditions trigger a clarification question rather than an immediate recommendation.
- Users can correct detected conditions, and partial matches are clearly identified.
- Recommendations are informational—not guarantees of plant survival, pet safety, or medical safety.

Fail-safe principle: If a required condition is missing or uncertain, ask the user instead of guessing.

Thank You!

